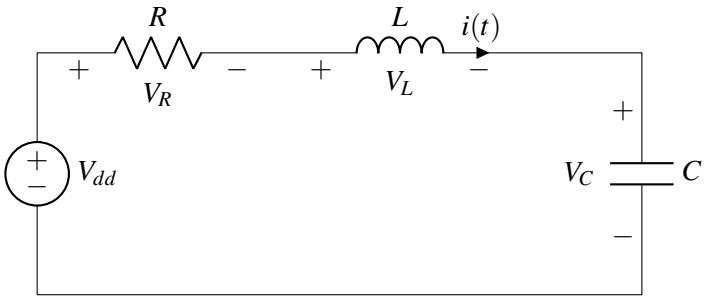
CSM
Spring 2026

EECS 16B
Worksheet 8

1. Introduction to Second Order Differential Equations

Take a RLC circuit in series. We will examine the behavior of this circuit with different R, L, and C values.



(a) Using your knowledge of I-V relationships of inductors and capacitors, write out a KVL equation that represents the behavior of this circuit.

(b) What is the steady state solution of $V_C(t)$?

(c) Solve for the solution of the homogenous differential equation.

(d) Let's analyze our solution. Under what conditions will we

i. Have two real exponent terms?

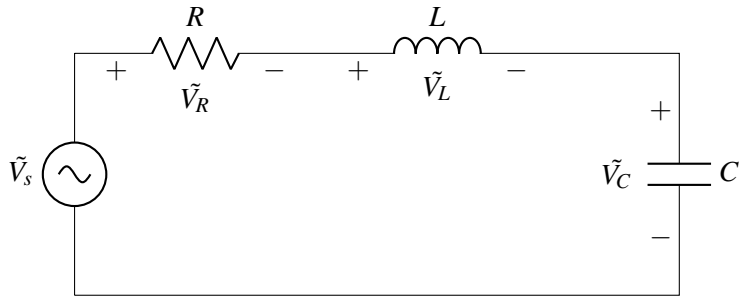
ii. Have only one exponent term?

iii. Have two complex exponent terms?

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2. RLC Circuit

In this question, we will take a look at an electrical systems described by second-order differential equations and analyze it in the phasor domain. Consider the circuit below where $\tilde{V}_s$ is a sinusoidal signal, $L = 1\,\text{mH}$, and $C = 1\,\text{nF}$:



(a) **Transform the circuit into the phasor domain.**

(b) **Solve for the transfer function $H_C(\omega) = \frac{\tilde{V}_C}{\tilde{V}_s}$ in terms of R, L, and C.**

(c) **Solve for the transfer function $H_L(\omega) = \frac{\tilde{V}_L}{\tilde{V}_s}$ in terms of R, L, and C.**

(d) **Solve for the transfer function $H_R(\omega) = \frac{\tilde{V}_R}{\tilde{V}_s}$ in terms of R, L, and C.**

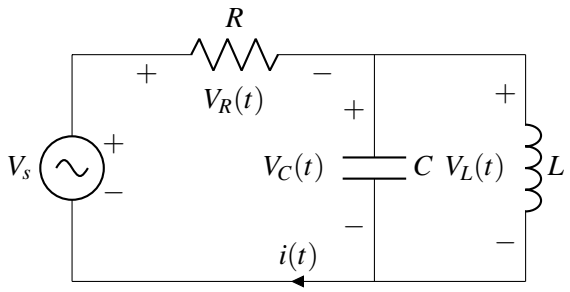
(e) **Sketch the Bode plots of $H_C(\omega)$, $H_L(\omega)$, and $H_R(\omega)$ when $R = 2\,\text{k}\Omega$.**

(f) **(Practice) Draw the Bode plot of $H_R(\omega)$ two more times, but change R to be $20\,\Omega$, then $200\,\text{k}\Omega$.**

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3. Power Analysis

Consider the following circuit:



$\omega = 1000\,\frac{\text{rad}}{\text{s}}$, $V_s(t) = 1000\sqrt{2}\cos\left(1000t + \frac{\pi}{2}\right)$, $R = 1000\,\Omega$, $C = 1\,\mu\text{F}$, $L = 0.5\,\text{H}$.

(a) First we need to transform the whole circuit from time domain to phasor domain. Please **redraw the circuit** with only $\widetilde{V}$, $\widetilde{I}$, and all impedances of the circuit components.

(b) **Find the phasor $\widetilde{I}$ through the circuit and the total impedance Z_{tot} of the circuit.**

(c) **Find the real power, reactive power, and apparent power.** *Hint:*

$$\text{apparent power} = \frac{1}{2}|\widetilde{V}||\widetilde{I}| \tag{1}$$

$$\text{reactive power} = \frac{1}{2}|\widetilde{V}||\widetilde{I}|\sin\left(\angle(Z_{tot})\right) \tag{2}$$

$$\text{real power} = \frac{1}{2}|\widetilde{V}||\widetilde{I}|\cos\left(\angle(Z_{tot})\right) \tag{3}$$

(d) Find the power factor, and state whether it is lagging or leading. *Hint:*

$$\text{power factor} = \frac{\text{real power}}{\text{apparent power}} = \cos(\theta_d),$$

where θ_d is phase difference betweent voltage and the current of the circuit.

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(e) Based on the previous question, **is the circuit's load resistive, inductive, or capacitive?**

Contributors:

- Jessica Fan, Olivia Huang.
- Kyle Tanghe, Damanic Luck.

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